Thomas Pratt

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PROFILE

Final-year Bachelor of Science student (Environmental Studies, Geology and Geophysics) at the University of Sydney, graduating 2026, with applied GIS and remote sensing experience: multi-criteria spatial modelling in ArcGIS Pro, satellite image classification and change detection in Google Earth Engine and land tenure analysis. Seeking a graduate role applying spatial analysis to the monitoring of nature-based projects.

TECHNICAL SKILLS

- **GIS:** ArcGIS Pro (ModelBuilder, multi-criteria analysis, raster and overlay analysis, cartography), QGIS
- **Remote Sensing:** Google Earth Engine, Sentinel-2 and Planet imagery, random forest classification, NDVI change detection and time series analysis
- **Programming:** Python (plate tectonics model simulation), JavaScript (Google Earth Engine)
- **Fieldwork:** Geological mapping, field sketching and stratigraphic logging (Orange and Bombo, NSW)
- **Policy familiarity:** EPBC Act 1999, Native Title Act 1993, NSW State Significant Development and Social Impact Assessment, ACCU Scheme

GIS AND REMOTE SENSING PROJECTS

Critical Minerals, Native Title and Social Impact Assessment, NSW **2026**
University of Sydney, group research project

- Contributed to GIS analysis of overlap between critical mineral developments/deposits and native title claims, alongside a review of NSW Social Impact Assessments for State Significant Developments.
- Conducted the literature review, synthesising 36 peer-reviewed sources on mining impacts under the Millennium Ecosystem Assessment framework.
- Found that 10 of 20 critical mineral mines in NSW lie on native title determination or registered claim land.

Jervis Bay Marine Park Zoning Plan **2026**
University of Sydney, five-person group project (GIS lead)

- Built the team's multi-criteria analysis model in ArcGIS Pro ModelBuilder to generate three alternative Sanctuary, Habitat and General Use zoning plans from a tourism stakeholder perspective.
- Combined six criteria layers (rocky reefs, seagrass and mangroves, commercial fishing grounds, wildlife buffers, boat ramp access and shipwrecks) with pairwise-weighted stakeholder priorities on a 10 m raster.
- Produced the final zoning maps and wrote the GIS methods section of the report.

Mangrove Dieback Analysis, Cabbage Tree Basin (Royal National Park) **2026**
University of Sydney, individual project

- Classified land cover (water, sand, saltmarsh, mangrove, trees) from Sentinel-2 imagery using a random forest model in Google Earth Engine, validated through accuracy assessment.
- Mapped and quantified mangrove dieback area by NDVI differencing of Planet imagery and identified the timing of dieback from a Sentinel-2 NDVI time series.

EDUCATION

University of Sydney **2024 - 2026**

- Bachelor of Science, majoring in Environmental Studies and minoring in Geology and Geophysics.

University of New South Wales **2023**

- Bachelor of Engineering (Honours), Software Engineering: completed first year (programming, algorithms, systems design) before transferring to environmental studies.

Knox Grammar School **2020 - 2022**

- Higher School Certificate, ATAR 95.05: ranked 1st in school in Earth and Environmental Science

WORK EXPERIENCE

Stihl Shop Stanmore | *Retail Assistant* **August 2024 - Present**

- Advise local council representatives and contractors on equipment purchases aligned with council maintenance and landscaping requirements.
- Recommend chainsaws, trimmers, blowers and safety gear matched to customer needs and provide post-purchase troubleshooting and maintenance advice.
- Open and close the store as a key holder and run the front of the store unsupervised.
- Train and supervise new staff on products, safety gear and the POS system.

The Athlete's Foot | *Retail Assistant* **March 2023 - April 2025**

- Produced sales reports and analysed product performance in MS Excel and Erply ERP.